Will Handley

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Summary

willhandley.co.uk/CV

Cosmologist and statistician whose research programme weaves theory, observation & inference: *Nested sampling; Bayesian machine learning; cosmological model selection, parameter estimation & tension quantification; likelihood-free inference; early universe cosmology; CMB; 21cm; gravitational waves; exoplanets.*

Education

2012–2016 **University of Cambridge**, *PhD Astrophysics*, Prof. A. Lasenby & Prof. M. Hobson

2008–2012 **University of Cambridge**, *MSci, MA: Natural Sciences*, Gonville & Caius College

2001–2008 **Alleyn's School**, *A levels, GCSEs*, London

Employment & Research

Oct 2024– **Associate Professor**, *Institute of Astronomy*, University of Cambridge

Oct 2020– **Royal Society University Research Fellow**, *IoA*, University of Cambridge
Bayesian machine learning and tensions in cosmology (Cavendish Lab 2020-2024)

May 2021– **Teaching Fellow**, *Gonville & Caius College*, University of Cambridge

2017– **Chief Technical Officer**, *PolyChord Ltd*, polychord.co.uk

2021–2023 **Turing Fellow**, *Alan Turing Institute*

2016–2020 **Research fellow**, *Gonville & Caius College*, University of Cambridge
Funded by Gonville & Caius College and an STFC IPS grant.

Jul-Sep 2016 **Postdoctoral researcher**, *Prof. H. Peiris*, University College London
Searching for features in the primordial power spectrum.

2012–2016 **PhD Astrophysics**, *Prof. A. Lasenby & Prof. M. Hobson*, University of Cambridge
Kinetic initial conditions for inflation: Theory, observations & methods.

2011–2012 **Part III Dissertation**, *Prof. P. Alexander*, University of Cambridge
Investigating the origins of cosmic magnetism.

Summer 2011 **Summer Research Student**, *Prof. M. Faulkes & Dr. J. Spencer*, Imperial College
Folded spectrum full configuration interaction quantum Monte Carlo.

Summer 2011 **Summer Research Student**, *Dr. R. Blumenfeld*, University of Cambridge
Geometry and field equations of granular systems.

2010–2011 **Research Review**, *Prof. S. Gull*, University of Cambridge
Literature Survey of the Physics-Philosophy crossover field of measurement theory.

Summer 2010 **iGEM Team Physicist**, *Dr. J. Haseloff*, University of Cambridge
E-glowli 2010 iGEM team (placed in final 6) http://2010.igem.org/Team:Cambridge

Grants won (£5m)

Cosmology

£690k **Royal Society URF Renewal 2025**, *Resolving the Universe with Scientific AI*

£1.3m **ERC starting grant ⇒ UKRI frontier research**, *Resolving cosmological tensions with diverse data, novel theories and Bayesian machine learning*
Oct 2024 willhandley.co.uk/ERC.pdf

£240k **Royal Society Enhancement 2023**, *Next generation nested sampling for cosmological inference*

£170k **Royal Society Enhancement 2021**, *Likelihood-free inference and Bayesian neural networks*

£723k **Royal Society URF 2020**, *Bayesian machine learning and tensions in cosmology*

16.1MCPUh ≡ £161k **DiRAC RAC 17th call 2025**, *New Horizons in Cosmology: Machine Learning and Inference for the Next Era of Data Analysis*

200kGPUh ≡ £80k **AIRR Early Access allocation**, *AInstein: a limited time-window trained LLM for (re)formulating modern physics*

52MCPUh ≡ £520k **DiRAC RAC 15th call 2023**, *New frontiers in particle cosmology*

30MCPUh ≡ £300k **DiRAC RAC 13th call 2021**, *Next generation cosmological analysis with nested sampling*

2MCPUh ≡ £20k **DiRAC directors discretionary 2020**, *Bayesian model comparison of inflation and spatial curvature*

£3k **Google Cloud compute award**, *GPU nested sampling research*

£15k **KICC Workshop 2023**, *GAMBIT at the KICC*

$6k **George Southgate Visiting Fellowship 2020**, *GAMBIT visit*

£15k **KICC Workshop 2019**, *AstroHackWeek 2019*

£2k **KICC visitors 2019**, *Likelihood free inference workshop*

£2k **KICC visitors 2017**, *Class and MontePython workshop*

PolyChord

£260k **MSCA DTN 2023**, *GLITTER: Gnss-r sateLlITe earTh obsERvation*, PC Ltd *et al*

£100k **DASA GAN 2023**, *Optimal dynamic manoeuvring & adaptation of communications networks*, PC Ltd

£25k **DSTL CEME 2023**, *MIDAS: Maximum information data acquisition strategies*, PC Ltd

£100k **DSTL CEME 2022**, *Further optimisation of sensor location*, PC Ltd & QML

£60k **DSTL CEME 2021**, *Optimisation of sensor location*, PC Ltd & QML

£10k **DSTL CEME 2020**, *Optimising search route for constrained network discovery*

£50k **Amadeus Seed capital**, *PolyChord for protein folding*, PC Ltd

£225k **STFC IPS 2019**, *PolyChord and Bayesian sparse facial recognition*

£42k **STFC IAA 2018**, *PolyChord and Bayesian neural network facial recognition*

£25k **STFC IAA 2016**, *Interfacing PolyChord 2.0*

Awards & Prizes

Jul. 2022	Pacific Institute of Theoretical Physics visitor	*University of British Columbia*
Feb. 2020	George Southgate visiting Fellow	*University of Adelaide*
Jul. 2019	Guiseppe and Vanna Cocconi Prize (WMAP and Planck)	*EPS-HEPP Division*
Dec. 2013	Best presentation	*Cavendish grad. students conference*
Jun. 2012	Best theoretical part III project	*University of Cambridge*
	Physics prize	*Gonville & Caius College*
Summer 2011	Undergraduate Research Bursary	*Nuffield Foundation*
	UROP Studentship	*Imperial College*
Summer 2010	iGEM Studentship	*Wellcome Trust*
2009–12	Junior and Senior Scholarships	*Gonville & Caius College*

Current students & postdocs

handley-lab.co.uk/group

Postdoc	Jiamin Hou, Chris Lovell, Natalie Hogg, Matt Grayling	*2025-present*
	David Yallup	*2021-present*
PhD	Charlotte Priestley	*2025-present*
	Toby Lovick	*2024-present*
	Namu Kroupa, Dily Ong, Sam Leeney	*2023-present*
	Metha Prathaban, Wei-Ning Deng, Sinah Legner	*2022-present*

Previous students & postdocs

PhD

Sep 2025 **Adam Ormondroyd**, *Bayesian methods in a flexible universe: sampling, tension and dark energy*
co-supervised with Mike Hobson and Anthony Lasenby

Aug 2024 **Kilian Scheutwinkel**, *Simulation-based Bayesian machine learning methods for Cosmology and beyond*
co-supervised with Eloy de Lera Acedo

Aug 2024 **George Carter**, *The Bayesian Global Sky Model (B-GSM)*
co-supervised with Nima Razavi-Ghods & Mark Ashdown

Mar 2024 **Thomas Gessey-Jones**, *Probing the First Stars with the 21-cm Signal: Theory, Methods, and Forecasts*
co-supervised with Eloy de Lera Acedo & Anastasia Fialkov

Dec 2023 **Ian Roque**, *EXCALIBRATE: Calibration for astrophysical experimentation*
co-supervised with Nima Razavi-Ghods

Jun 2023 **Harry Bevins**, *A Machine Learning-enhanced Toolbox for Bayesian 21-cm Data Analysis and Constraints on the Astrophysics of the Early Universe*
co-supervised with Eloy de Lera Acedo and Anastasia Fialkov

Jul 2022 **Dominic Anstey**, *Data Analysis in Global 21cm Experiments: Physically Motivated Bayesian Modelling Techniques*
co-supervised with Eloy de Lera Acedo

Sep 2021 **Fruzsina Agocs**, *Primordial evolution of cosmological perturbations: Theory and computation*
co-supervised with Mike Hobson & Anthony Lasenby

Aug 2021 **Will Barker**, *Gauge Theories of Gravity*
co-supervised with Mike Hobson & Anthony Lasenby

Dec 2020 **Lukas Hergt**, *Constraining the kinetically dominated Universe*
co-supervised with Mike Hobson & Anthony Lasenby

Oct 2018 **Ed Higson**, *Bayesian Methods and machine Learning in Astrophysics*
co-supervised final year with Mike Hobson & Anthony Lasenby

Individual projects Thomas McAloone (2020-21), Isidro Gómez Vargas (2020), Ayngaran Thavenesan (2021-22)

MPhil

Sep 2023 **Danielle Dineen**, *Cosmological Matching Conditions for Primordial Perturbations*

Jan 2023 **Allahyar Sahibzada**, *Machine Learning and Nested Sampling: in the context of data intensive science and cosmology*

Nov 2022 **Sam Leeney**, *Data science in early universe Cosmology: a novel Bayesian RFI mitigation approach using numerical sampling techniques*
co-supervised with Eloy de Lera Acedo

Aug 2020 **Emma Shen**, *Ionospheric Effects in the Global 21-cm Experiment*
co-supervised with Eloy de Lera Acedo & Anastasia Fialkov

Aug 2019 **Ian Roque**, *Bayesian Techniques for the Calibration of 21 cm Global Experiments*
co-supervised with Nima Razavi-Ghods

Aug 2018 **Panagiotis Mavrogiannis**, *Wheeler–Feynman absorber theory of radiation: Establishing the cosmological electrodynamic arrow of time*
co-supervised with Anthony Lasenby

MSci

2025 Harvey Williams, Krish Nanavati, Ming Yang, Will Templeton, Charlotte Priestley
2024 Nicolas Mediato Diaz, Samuel Hewson, Felicity Ibrahim, Patrick Lau, Tze Goh
2023 Zixiao Hu, Cole Meldorf, Sankalan Bhattacharyyan, Toby Lovick
2022 Yoann Launay, Oliver Normand, Xy Wang, Carola Zanoletti
2021 Yi Jer Loh, Metha Prathaban
2020 Thomas Gessey-Jones, Aleks Petrosyan
2019 Deaglan Bartlet, Jamie Bamber, Ian Roque
2018 Ward Haddadin, Jessica Rigley
2017 Fruzsina Agocs, Robert Knighton, Stephen Pickman, Daniel Manela

Summer students

2024 Charlotte Priestley
2023 Zixiao Hu, Toby Lovick, Namu Kroupa
2022 Mary Letey, Beichen Xu, Artyom Baryshnikov
2021 Zak Shumaylov, Mattia Varrone
2019 Denis Werth, Maxime Jabarian, Liam Lau
2018 Elizabeth Guest, Ward Haddadin, Shu-Fan Chen

Postdocs

2024 Thomas Gessey-Jones
2020 Jianghui Lui
2018-19 Kamran Javid

Lecturing

2021-2023 Part III Physics: Relativistic Astrophysics & Cosmology *MSci 24 lecture course*
2017-2021 Bayesian Statistics *Graduate 2 lecture course*

Workshops

2025 **GPU-native nested sampling in BlackJAX**, *For simulation-based inference at scale*, SBI Galaxy Evolution 2025, Bristol
github.com/handley-lab/workshop-blackjax-nested-sampling

2025 **AI/ML Tools for Research**, *KICC Workshop*, July 2025

2025 **An Introduction to Artificial Intelligence and Large Language Models**, *Part II/III students*, February 2025
docs.google.com/presentation

2023 **Monte Carlo Methods**, *For Cosmology and Particle Physics*, UNAM, Mexico
github.com/handley-lab/workshop-monte-carlo-methods

2022 **ICCS**, *Training Machine Learning models*, Cambridge, UK
github.com/handley-lab/2022-cambridge-iccs

2018 **CosmoTools**, *Introduction to Statistics*, Aachen, Germany
indico.cern.ch/e/CosmoTools2018

2017 **CosmoTools**, *Cosmological statistics & sampling*, IFT Madrid, Spain
workshops.ift.uam-csic.es/cosmotools2017

Small group teaching

2020–present	Part III Physics: Relativistic Astrophysics and Cosmology	*Supervising (24 hours)*
2013–present	Part II Physics: General relativity	*Supervising (156 hours)*
2023–present	Part II Physics: Statistical Mechanics	*Supervising (28 hours)*
2012–2017	Part IA Mathematics for NatSci	*Tripos classes (20 hours),Supervising (580 hours)*
2015-2016	Part IA Physics	*Supervising (20 hours)*
2013	Part II Theoretical Physics 1 & 2	*Demonstrating (8 hours)*
2006–2012	Maths and Science Tuition	*Individual coaching, key stage 1 – STEP*

Academic Talks willhandley.co.uk/talks † = remote

Sep. 2025 **Next-Generation Model Comparison for Primordial Cosmology**, *LMU Munich*, Munich, Germany

Jul. 2025 **A Statistician's Guide to the Galaxy (Fitting Zoo)**, *Future of SED Fitting Workshop*, Cambridge, UK

Jun. 2025 **GPU Accelerated Nested Sampling**, *Newton Institute workshop*, Cambridge, UK

Jun. 2025 **lsbi: linear simulation based inference**, *EUCAIFCON 2025*, Sardinia

May. 2025 **GPU-native nested sampling in BlackJAX: For simulation-based inference at scale**, *SBI Galaxy Evolution 2025*, Bristol

May. 2025 **Scanning for cosmological tensions across a DiRAC-enabled grid of models, datasets and samplers**, *Cosmoverse 2025*, Italy

Jan. 2025 **Cosmological tensions? A guide for high energy theorists**, *DAMTP HEP group*, Cambridge, UK

Jan. 2025 **Theory meets experiment 2025: New frontiers in particle cosmology**, *Rencontres du Vietnam*, Quy Nhon, Vietnam

Dec. 2024 **Next-generation statistical inference tools: Simulation-based inference, marginal statistics & accelerated nested sampling**, *Towards a realistic detection of Primordial Gravitational Wave Backgrounds*, Madrid, Spain

Oct. 2024 **PolySwyft: a sequential simulation-based nested sampler**, *Global 21cm workshop 2024*, Raman Research Institute, Bangalore, India

Sep. 2024 `lsbi`: **linear simulation based inference**, *PhyStat: Statistics meets ML*, Imperial college London, UK

Aug. 2024 **Nested sampling: powering next-generation inference and machine learning tools for astrophysics, cosmology, particle physics and beyond**, *University of Sydney*, Sydney, Australia

Aug. 2024 **Sampling methods for high energy physics & particle astrophysics**, *XVIth Quark Confinement and the Hadron Spectrum*, Cairns, Australia

Aug. 2024 **Nested sampling: powering next-generation inference and machine learning tools for astrophysics, cosmology, particle physics and beyond**, *University of Queensland*, Brisbane, Australia

Jul. 2024 **Nested sampling: powering next-generation inference and machine learning tools for astrophysics, cosmology, particle physics and beyond**, *RWTH*, Aachen, Germany

Jul. 2024 **PolySwyft: a sequential simulation-based nested sampler**, *CosmoVerse 2024*, Krakow, Poland

Jul. 2024 **The scaling frontier of nested sampling**, *MaxEnt 2024*, Ghent, Belgium

Jun. 2024 **Resonant or asymmetric: The status of sub-GeV dark matter**, *Dark Matter in Astrophysical Laboratories*, Cambridge, UK

May. 2024 **Next generation astrophysical inference across the interdisciplinary frontier**, *UCL job talk*, UCL, UK†

May. 2024 **PolySwyft: a sequential simulation-based nested sampler**, *PhyStat 2024*, CERN, Switzerland

Apr. 2024 **Next generation astrophysical inference across the interdisciplinary frontier**, *IoA job talk*, Cambridge, UK

Mar. 2024 **Nested sampling: powering next-generation inference and machine learning tools for astrophysics, cosmology, particle physics and beyond**, *Gatsby computational neuroscience unit*, UCL, UK

Feb. 2024 **Sampling techniques in high-dimensional parameter spaces with ScannerBit 2.0**, *ORIGINS data science cluster*, Munich, Germany

Jan. 2024 **Next-generation inference tools for cosmology and beyond**, Oxford, UK

Jan. 2024 **Simulation Based Inference: theory, sampling & model comparison**, *RAS*, London, UK

Dec. 2023 **Nested sampling: powering next-generation inference and machine learning tools for astrophysics, cosmology, particle physics and beyond**, *UNAM*, Mexico City, Mexico

Nov. 2023 **Bayesian OODA loops with MIDAS: Augmented decision making in a complex future electromagnetic environment**, *OFEME 2023*, Nottingham, UK

Oct. 2023 **unimpeded: Universal model comparison and parameter estimation distributed over every dataset**, *Oscar Klein Center*, Stockholm, Sweden

Oct. 2023 **Nested sampling: powering next-generation inference and machine learning tools for cosmology, particle physics and beyond**, *Manchester*, UK

Sep. 2023 **Nested sampling tools**, *REACH AGM*, Malta

Sep. 2023 **Nested sampling: powering next-generation inference and machine learning tools for cosmology, particle physics and beyond**, *UCL*, UK

Aug. 2023 **Nested sampling: powering next-generation inference and machine learning tools for cosmology, particle physics and beyond**, *KCL*, UK

Jul. 2023 **The scaling frontier of nested sampling: Summary talk**, *MaxEnt*, Munich, Germany

Jun. 2023 **Gradients and Nested Sampling: the present state of the art**, *MIAPbP*, Munich, Germany

Mar. 2023 **Nested Sampling: A multi-purpose numerical tool for science and machine learning**, ETH Zurich, Switzerland

Jan. 2023 **Nested sampling: powering the next-generation of Bayesian inference tools for cosmology, particle physics and beyond**, *Cavendish job talk*, Cambridge, UK

Jan. 2023 **High dimensional nested sampling**, *Simulation based inference with swyft*, Amsterdam, Netherlands

Jan. 2023 **What is the benefit of adversarial systems?**, *Mathematical Challenges in the Electromagnetic Environment*, London, UK

Dec. 2022 **Theory, observation & cosmological inference**, *KICC christmas*, Cambridge, UK

Sep. 2022 **Next generation cosmological analysis with nested sampling**, *KICC Symposium*, Cambridge, UK

Sep. 2022 **Next generation cosmological analysis with nested sampling**, *Corfu2022: Tensions in Cosmology*, Corfu, Greece

Aug. 2022 **Dark matter, cosmology and likelihood-free Inference**, *GAMBIT XIV*, Kelowna, Canada

Jul. 2022 **Nested Sampling: An efficient and robust Bayesian inference tool for particle physics and cosmology**, *TRIUMF & UBC*, Vancouver, Canada

Jul. 2022 **Frontiers of Nested Sampling**, *MaxEnt 2022*, Paris, France

Apr. 2022 **Nested Sampling and Likelihood-free inference**, *Likelihood-free in Paris*, Paris, France

Apr. 2022 **Statistical methods in Cosmology**, *Obs. and Theor. 21-cm Cosmology*, Cambridge, UK

Jan. 2022 **PolyChord: Next generation nested sampling**, *UK Atomic Energy Authority*, UK†

Nov. 2021 **Review on Statistical Tools and Samplers**, *TOOLS 2021*, IP2I, Lyon, France†

Jul. 2021 **Success Story 2 — Optimum Sensor Placement**, *Mathematical Challenges in the Electromagnetic Environment*, Isaac Newton Institute, Cambridge, UK

Jul. 2021 **Success Story 1 — Detecting Illicit Mesh Networks**, *Mathematical Challenges in the Electromagnetic Environment*, Isaac Newton Institute, Cambridge, UK

Mar. 2021 **PolyChord: Novel Bayesian Machine Learning**, *Cambridge Data Science Fair*, UK†

Feb. 2021 **Bayesian methods for quantifying global parameter tensions between cosmological datasets**, *Tehran meeting on cosmology at the crossroads*, Tehran, Iran†

Jan. 2021 **Bayesian information fusion**, *Mathematical Challenges in the Electromagnetic Environment*, Isaac Newton Institute, Cambridge, UK†

Oct. 2020 **Nested Sampling: an efficient and robust Bayesian inference tool for 21cm cosmology**, *3rd Global 21-cm Workshop*, Cambridge, UK†

Sep. 2020 **Nested Sampling for optimising sensor location**, *Mathematical Challenges in the Electromagnetic Environment*, Isaac Newton Institute, Cambridge, UK†

Feb. 2020 **Nested Sampling: an efficient and robust Bayesian inference tool for physics and machine learning**, *Physics Colloquium*, Adelaide, Australia

Jan. 2020 **Nested Sampling: an efficient and robust Bayesian inference tool for astrophysics and cosmology**, Oxford, UK

Jan. 2020 **PolyChord: next generation nested sampling**, *Mathematical Challenges in the Electromagnetic Environment*, Isaac Newton Institute, Cambridge, UK

Dec. 2019 **Quantised primordial power spectra**, *Texas 2019*, Portsmouth, UK

Nov. 2019 **Nested Sampling: an efficient and robust Bayesian inference tool for Machine Learning and Data Science**, *CDT talk*, Cambridge, UK

Aug. 2019 **Curvature tension: evidence for a closed universe(?)**, ICG Portsmouth, UK

Jul. 2019 **Quantifying cosmological tensions**, University College London, UK

Jun. 2019 **Likelihood free inference**, *GAMBIT X*, Germany

Mar. 2019 **Compromise-free Bayesian sparse reconstruction**, *LFI workshop*, Flatiron institute, US

Dec. 2018 **Inflation, curvature and kinetic dominance**, *Future uses of Planck data*, ESAC, Spain

Nov. 2018 **BAMBI Resurrection: Blind Accelerated Multimodal Bayesian Inference**, *Dark Machines*, Worldwide†

Nov. 2018 **Nested Sampling: an efficient and robust Bayesian inference tool for cosmology and particle physics**, *Dark Machines*, Worldwide†

Oct. 2018 **Bayesian Statistics**, *Third Asterics-Obelics workshop*, Cambridge, UK

May. 2018 **Planck, inflation and the future of inflationary constraints**, *Consistency of Cosmological Datasets*, Cambridge, UK

May. 2018 **MaxEnt priors with derived parameters in a specified distribution**, Cambridge, UK

May. 2018 **Nested Sampling: an efficient and robust Bayesian inference tool for astrophysics and cosmology**, ICIC, UK

April. 2018 **Introduction to statistics**, *CosmoTools 18*, RWTH Aachen, Germany

Jan. 2018 **Advances in Nested Sampling & astrophysical application**, Cambridge, UK

Aug. 2017 **PolyChord 2.0: Fast inference & nested sampling**, *Cosmo17*, Paris, France

Jun. 2017 **Modern Bayesian Inference: Theory and Practice**, RWTH Aachen, Germany

Mar. 2017 **Parameter estimation and Model comparison**, *CosmoTools 17*, Madrid, Spain

Feb. 2017 **PolyChord 2.0: Advances in Nested Sampling & astrophysical application**, Flatiron institute, US

Sep. 2016 **PolyChord 2.0 & the future of nested sampling**, University College London, UK

May. 2016 **PolyChord 2.0 & the future of nested sampling**, University of Sussex, UK

Mar. 2016 **PolyChord & the future of nested sampling**, Edinburgh, UK

Dec. 2015 **PolyChord: next generation nested sampling**, *MPA Bayes Forum*, Munich, Germany

Feb. 2015 **PolyChord: next generation nested sampling**, University of Sussex, UK

Dec. 2013 **Kinetic dominance in the pre-inflationary universe**, Cavendish grad. conference

Outreach

17 Feb 2025 **Precision Cosmology: new scientific windows on the Universe**, *Sherrington Society*, Gonville & Caius College

24 Oct 2023 **Tensions in the Universe: a crisis in precision cosmology?**, *Thomas Young Club*, Emmanuel College

11 Feb 2019 **Cosmology: A Scientist's History of the Universe**, *Sherrington Society*, Gonville & Caius College

21 Sep 2018 **Observing the big bang: A short history of nearly nothing**, *Public talk*, Belgium

18 May 2016 **Cosmology**, *Intro to Astronomy course*, Cambridge

1 Jul 2015 **The Physics of Juggling**, *Senior Physics Challenge*, Churchill College

27 May 2015 **Intro. to Astronomy: Beyond the Milky Way**, *IoA Public Talk*, Cambridge

20 May 2015 **To infinity and beyond: Dark Energy**, *Pint of Science*, Cambridge Brewhouse

14 Jan 2015 **The Cosmic Microwave Background: our probe of the birth of the universe**, *IoA Public Evening*, Cambridge

26 Nov 2014 **Dark Matter and Dark Energy: the composition and fate of the universe**, *Letchworth & District Astronomical Society*, Letchworth

5 Oct 2014 **The Physics of Juggling**, *Fun Palaces*, Brockwell Lido, London

30 Jun 2014 **The Physics of Juggling**, *Senior Physics Challenge*, Corpus Christi College

28 May 2014 **Galaxies and beyond**, *Intro to Astronomy course*, Cambridge

19 Mar 2014 **Inflation: extreme expansion in the very earliest universe**, *IoA Public Evening*, Cambridge

10 Feb 2014 **The Physics of Juggling**, *CCPE*, Cavendish Laboratory

30 Jan 2014 **The first 3 yocto-pico seconds**, *Three minute wonder*, Cavendish Laboratory

18 Dec 2013 **Special Relativity**, *Hills Road Sixth Form College*, Cambridge

14 Oct 2013 **The Cosmic Microwave Background: our probe of the birth of the universe**, *IoA Public Evening*, Cambridge

2 Oct 2013 **Dark matter and dark energy: common misconceptions**, *IoA Public Evening*, Cambridge

8 Jul 2013 **Introduction to Cosmology**, *Hills Road Sixth Form College*, Cambridge

20 Mar 2013 **The Search for Dark Matter**, *NSEW Talk, Luton Astronomical Society*, Putteridge Bury

26 Sep 2012 **Cosmology**, *Alleyn's School*, London

Leadership development

Mar.–Sep. 2025 **Cambridge Leading Researchers Programme**, *Competitively selected pilot cohort*, 64 Million Artists & University of Cambridge
Multi-institutional leadership development programme with workshops and individual coaching

2017 **Impulse**, *STFC-funded entrepreneurship programme*, Cambridge Enterprise & Maxwell Centre
Technology commercialization training with mentoring, IP guidance, and grant application support

Institutional responsibilities

2025–present	Chair of AI working group	*Gonville & Caius college*
2024–present	Teaching committee	*Institute of Astronomy*
2024–present	CPAC (CATAM) committee	*University of Cambridge*
2024	CASU steering committee	*Institute of Astronomy*
2020–present	Convener of CosmoBit	*GAMBIT*
2020–2024	Leader of data analysis team	*REACH*
2021–present	Center for data-driven discovery (C2D3) steering committee	*University of Cambridge*
2025–present	CamCEAD Strategy Committee	*University of Cambridge*
2022–present	KICC Scientific Strategy Committee	*KICC*
2024–present	KICC Fellowships committee	*KICC*
2022–present	KICC Visitor and Lecturer committee	*KICC*
2018–present	Science Research Fellowships committee	*Gonville & Caius college*
2018–present	Investments committee	*Gonville & Caius college*
2016–present	Undergraduate Admissions	*Gonville & Caius college*
2020–present	Wine Committee (WSET3)	*Gonville & Caius college*
2019–2022, 2024–present	Gonville & Caius College Council	*Gonville & Caius college*
2021–2023	CDT in data intensive science executive committee	*University of Cambridge*
2018–2020	Education and research committee	*Gonville & Caius college*
2017–2022	Organiser of weekly group seminars	*Cavendish astrophysics group*

Examination

2024– **Senior Examiner**, *Astrophysics*, Part II

2024– **CATAM coordinator**, *Astrophysics*, Part II

2021–2023 **Exam setting**, *Relativistic Astrophysics and Cosmology*, Part III Physics

2020–2022 **Masters exam checking**, *Astrostatistics*, Part III Maths

PhD

2025 **Emilie Hertig**, *Probes of cosmic inflation: from the CMB to quantum analogues*

Jul 2025 **Matthew Craigie**, *Interpretable and Physically-Motivated Deep Learning Solutions for Large-Scale Structure Cosmology*

May 2024 **Stefan Heimersheim**, *Constraining reionization: Evidence from 21 cm limits and predictions for fast radio bursts*

Dec 2023 **Lester Sandles**, *Star-forming Galaxies and Quenched Systems throughout Cosmic Time*

Dec 2021 **Wu Hyun Sohn**, *High-resolution CMB bispectrum estimator*

First year reports (probationary reviews)

Sep 2025 **Mohammad-Hadi Sotoudeh**

Aug 2025 **Edward Stevenson**
Nov 2023 **Yuchen Liu**
Mar 2023 **Dily Ong**
Aug 2022 **Yu Hsuan Shen**

Second year reports

Jul 2025 **Alexander Byrne**

MPhil

Sep 2020 **Kyriakos Stylianiopoulos**, *Machine Learning Applied to Gaia and Other Survey Data: Applications Supporting a Polarisation Survey*

Organisation of scientific meetings

2025	LoC member forIOP Joint APP and HEPP Annual Conference 2025	*Cavendish*
2024	Cosmological Inference in High Dimension	*KICC*
2023	GAMBIT at the KICC	*KICC*
2023	Frontiers of nested sampling	*Munich*
2023	Simulation based inference workshop (delayed from 2019)	*KICC*
2020	Scientific organising committee member of 3rd Global 21-cm Workshop	*KICC*
2019	Local organising committee member of KICC 10th anniversary symposium	*KICC*
2019	Helped secure funding and organised AstroHack week 2019	*KICC*
2018	Secured funding for and organised CLASS+MontePython software workshop	*KICC*

Peer review

Performed 78 reviews for journals including Physical Review D and Physical Review Letters;
`https://www.webofscience.com/wos/author/record/S-9134-2018`
PRD (34), MNRAS (7), JCAP (8), PRL (8), JOSS (2), APJ (2), EPJC (1), PLB (6), RASTI (2) Entropy (4), Astronomy & Computing (2), Physics of the Dark Universe (2)

Review for fellowship awards:

2022 C2D3 Early Career Researcher Seed Fund
2022 ABTA UK Doctoral Research Award
2022 Blavatnik fellowship
2021– Gonville & Caius Junior Research Fellowships

Collaborations

2018–present	REACH	*astro.phy.cam.ac.uk/research/research-projects/reach*
2018–present	GAMBIT	*gambit.hepforge.org*
2018–2020	DarkMachines	*darkmachines.org*
2017–2018	Terra Hunter Experiment	*terrahunting.org*
2016–2017	CORE	*core-mission.org*
2015–2016	AMI	*astro.phy.cam.ac.uk/research/research-projects/AMI*
2015–2019	Planck	*cosmos.esa.int/web/planck*

Software willhandley.co.uk/software

PolyChord Sole author and maintainer: github.com/PolyChord/PolyChordLite
anesthetic Principle author and maintainer: github.com/handley-lab/anesthetic

mcp-handley-lab	Principle author and maintainer: github.com/handley-lab/mcp-handley-lab
blackjax	Contributor (nested sampling): github.com/handley-lab/blackjax
lsbi	Principle author and maintainer: github.com/handley-lab/lsbi
unimpeded	Principle author and maintainer: github.com/handley-lab/unimpeded
fgivenx	Sole author and maintainer: github.com/handley-lab/fgivenx
pyBAMBI	Team maintainer: github.com/DarkMachines/pyBAMBI
MultiNest	Maintainer: github.com/farhanferoz/MultiNest
primordial	Sole author and maintainer: github.com/williamjameshandley/primordial
ModeCode	Maintainer: modecode.org
Open source	scipy: Weighted kernel density estimation in scipy.stats.gaussian_kde matplotlib: Vertical slider in matplotlib.widgets.Slider

Interaction with industry

PolyChord	2017–: Founded start-up company PolyChord Ltd. to bring Bayesian methods & tools from cosmology to Machine Learning & Biotech industries: polychord.co.uk
CMAM	2017–2023: Consulted for finance spin-out on Bayesian algorithmic trading
Shell	2016: Worked with department postdocs in the department applying nested sampling to geophysics
AnyVision	2019–2020: Worked collaboratively as part of STFC grant to apply Bayesian sparse reconstruction to facial recognition

Interaction with Government

2025	Invited participant: "11 Years of Resistance: Advancing the UK–Ukraine Strategic Partnership" conference, Houses of Parliament
2025	AI consult to the UK Technology Advisor (Dave Smith)
2024	Workshop participant in national security resilience in the future electromagnetic environment
2020–	DSTL: Consult for government defence research using Bayesian inference

In the media

2022 **Cavendish Laboratory News**, *What can astrophysical data-intensive science do beyond the Universe?*, PolyChord, the next generation optimisation technology
https://www.phy.cam.ac.uk/news/what-can-astrophysical-data-intensive-science-do-beyond-universe-polychord-next-generation

2022 **BBC Radio 4**, *The Third Degree*, Astrophysics Don
"Students vs Dons" BBC radio quiz aired July 2022

2022 **KICC annual report**, *Bringing astrostatistics back to Earth*
kicc.cam.ac.uk/aboutus/kicc-annual-reports

2020 **Quanta Magazine**, *Modified gravity in cosmology led by Will Barker*
quantamagazine.org/why-is-the-universe-expanding-so-fast-20200427/

2019 **KICC annual report**, *Compromise-free Bayesian cosmology & AstroHack week*

Computer skills

Programming	MPI parallelisation, C++, FORTRAN, Mathematica, Maple, Python, JAX
Computing	Unix, Bash, zsh, vim, git, svn, LaTeX, Ti*k*Z, VMs, CI, LLMs (Claude, Gemini, GPT)
OS	Arch Linux & HPC supercomputing (Experienced), Windows & OSX (Familiar)

References

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Publications: arxiv.org/a/handley_w_1

ui.adsabs.harvard.edu/search/q=orcid%3A0000-0002-5866-0445

[1] Íñigo Zubeldia, Boris Bolliet, Anthony Challinor, and **William Handley**. Extracting cosmological information from the abundance of galaxy clusters with simulation-based inference. *arXiv*, 2504.10230, April 2025.

[2] David Yallup, Metha Prathaban, James Alvey, and **Will Handley**. Parallel Nested Slice Sampling for Gravitational Wave Parameter Estimation. *arXiv*, 2509.24949, September 2025.

[3] Peter H. Sims, Harry T. J. Bevins, Anastasia Fialkov, Dominic Anstey, **Will J. Handley**, Stefan Heimersheim, Eloy de Lera Acedo, Rajesh Mondal, and Rennan Barkana. Rapid and Late Cosmic Reionization Driven by Massive Galaxies: a Joint Analysis of Constraints from 21-cm, Lyman Line & CMB Data Sets. *arXiv*, 2504.09725, April 2025.

[4] REACH Collaboration. Receiver design for the REACH global 21-cm signal experiment. *Experimental Astronomy*, 59(1):7, February 2025.

[5] REACH Collaboration. Radiometer Calibration using Machine Learning. *arXiv*, 2504.16791, April 2025.

[6] Metha Prathaban, David Yallup, James Alvey, Ming Yang, Will Templeton, and **Will Handley**. Gravitational-wave inference at GPU speed: A bilby-like nested sampling kernel within blackjax-ns. *arXiv*, 2509.04336, September 2025.

[7] Metha Prathaban, Harry Bevins, and **Will Handley**. Accelerated nested sampling with posterior repartitioning and β-flows for gravitational waves. *MNRAS*, 541(1):200–213, July 2025.

[8] A. N. Ormondroyd, **W. J. Handley**, M. P. Hobson, and A. N. Lasenby. Non-parametric reconstructions of dynamical dark energy via flexknots. *MNRAS*, 541(4):3388–3400, August 2025.

[9] A. N. Ormondroyd, **W. J. Handley**, M. P. Hobson, and A. N. Lasenby. Dynamic or Systematic? Bayesian model selection between dark energy and supernova biases. *arXiv*, 2509.13220, September 2025.

[10] A. N. Ormondroyd, **W. J. Handley**, M. P. Hobson, and A. N. Lasenby. Comparison of dynamical dark energy with ΛCDM in light of DESI DR2. *arXiv*, 2503.17342, March 2025.

[11] Nicolas Mediato-Diaz and **Will Handley**. Cosmological Parameter Estimation with Sequential Linear Simulation-based Inference. *arXiv*, 2501.03921, January 2025.

[12] Toby Lovick, David Yallup, Davide Piras, Alessio Spurio Mancini, and **Will Handley**. High-Dimensional Bayesian Model Comparison in Cosmology with GPU-accelerated Nested Sampling and Neural Emulators. *arXiv*, 2509.13307, September 2025.

[13] Toby Lovick, Suhail Dhawan, and **Will Handley**. Non-Gaussian likelihoods for Type Ia supernova cosmology: implications for dark energy and H_0. *MNRAS*, 536(1):234–246, January 2025.

[14] Pablo Lemos, Nikolay Malkin, **Will Handley**, Yoshua Bengio, Yashar Hezaveh, and Laurence Perreault-Levasseur. Improving Gradient-guided Nested Sampling for Posterior Inference. *Proceedings of Machine Learning Research*, 235:27230–27253, January 2025.

[15] Sinah Legner, **Will Handley**, Will Barker, and Adam Ormondroyd. Alleviating the Hubble tension with Torsion Condensation (TorC). *arXiv*, 2507.09228, July 2025.

[16] S. A. K. Leeney, **W. J. Handley**, H. T. J. Bevins, and E. de Lera Acedo. Bayesian Anomaly Detection for Ia Cosmology: Automating SALT3 Data Curation. *arXiv*, 2509.13394, September 2025.

[17] Namu Kroupa, Gábor Csányi, and **Will Handley**. Resonances in reflective Hamiltonian Monte Carlo. *arXiv*, 2504.12374, April 2025.

[18] Namu Kroupa, Gábor Csányi, and **Will Handley**. Resonances in reflective Hamiltonian Monte Carlo. *PRE*, 111(4):045308, April 2025.

[19] Samuel Hewson, **Will Handley**, and Christopher G. Lester. Spatial distribution of the large-scale structure: An unsupervised search for parity violation. *PRD*, 111(12):123528, June 2025.

[20] T. Gessey-Jones, N. S. Sartorio, H. T. J. Bevins, A. Fialkov, **W. J. Handley**, E. de Lera Acedo, G. M. Mirouh, R. G. Izzard, and R. Barkana. Determination of the mass distribution of the first stars from the 21-cm signal. *Nature Astronomy*, 9:1268–1279, August 2025.

[21] Wei-Ning Deng and **Will Handley**. CMB Constraints on Quantized Spatial Curvature Ω_K in globally CPT-symmetric universes. *arXiv*, 2509.10379, September 2025.

[22] CosmoVerse. The CosmoVerse White Paper: Addressing observational tensions in cosmology with systematics and fundamental physics. *Physics of the Dark Universe*, 49:101965, September 2025.

[23] George Carter, **Will Handley**, Mark Ashdown, and Nima Razavi-Ghods. The Bayesian Global Sky Model (B-GSM): Validation of a Data Driven Bayesian Simultaneous Component Separation and Calibration Algorithm for EoR Foreground Modelling. *arXiv*, 2501.01417, January 2025.

[24] George Carter, **Will Handley**, Mark Ashdown, and Nima Razavi-Ghods. The Bayesian Global Sky Model (B-GSM): A Calibrated Low Frequency Sky Model for EoR Applications. *arXiv*, 2504.04503, April 2025.

[25] Harry T. J. Bevins, **William J. Handley**, and Thomas Gessey-Jones. Calibrating Bayesian tension statistics using neural ratio estimators. *PRD*, 112(4):043520, August 2025.

[26] H. T. J. Bevins, T. Gessey-Jones, and **W. J. Handley**. On the accuracy of posterior recovery with neural network emulators. *MNRAS*, September 2025.

[27] Sowmiya Balan, Csaba Balázs, Torsten Bringmann, Christopher Cappiello, Riccardo Catena, Timon Emken, Tomás E. Gonzalo, Taylor R. Gray, **Will Handley**, Quan Huynh, Felix Kahlhoefer, and Aaron C. Vincent. Resonant or asymmetric: the status of sub-GeV dark matter. *JCAP*, 2025(1):053, January 2025.

[28] Joshua Albert, Csaba Balázs, Andrew Fowlie, **Will Handley**, Nicholas Hunt-Smith, Roberto Ruiz de Austri, and Martin White. A comparison of Bayesian sampling algorithms for high-dimensional particle physics and cosmology applications. *Computer Physics Communications*, 315:109756, October 2025.
[29] Margret Westerkamp, Jakob Roth, Philipp Frank, **Will Handley**, and Torsten Enßlin. Towards a Field-Based Bayesian Evidence Inference from Nested Sampling Data. *Entropy*, 26(11):930, October 2024.
[30] Anchal Saxena, P. Daniel Meerburg, Christoph Weniger, Eloy de Lera Acedo, and **Will Handley**. Simulation-based inference of the sky-averaged 21-cm signal from CD-EoR with REACH. *RAS Techniques and Instruments*, 3(1):724–736, January 2024.
[31] Metha Prathaban and **Will Handley**. Costless correction of chain based nested sampling parameter estimation in gravitational wave data and beyond. *MNRAS*, 533(2):1839–1851, September 2024.
[32] Michael Pagano, Peter Sims, Adrian Liu, Dominic Anstey, **Will Handley**, and Eloy de Lera Acedo. A general Bayesian framework to account for foreground map errors in global 21-cm experiments. *MNRAS*, 527(3):5649–5667, January 2024.
[33] M. I. Letey, Z. Shumaylov, F. J. Agocs, **W. J. Handley**, M. P. Hobson, and A. N. Lasenby. Quantum initial conditions for curved inflating universes. *PRD*, 109(12):123502, June 2024.
[34] Namu Kroupa, David Yallup, **Will Handley**, and Michael Hobson. Kernel-, mean-, and noise-marginalized Gaussian processes for exoplanet transits and H_0 inference. *MNRAS*, 528(2):1232–1248, February 2024.
[35] Christian J. Kirkham, **William J. Handley**, Jiacong Zhu, Kaan Artuc, Ian L. V. Roque, Samuel A. K. Leeney, Harry T. J. Bevins, Dominic J. Anstey, and Eloy de Lera Acedo. Accounting for Noise and Singularities in Bayesian Calibration Methods for Global 21-cm Cosmology Experiments. *arXiv*, 2412.14023, December 2024.
[36] Zixiao Hu, Artem Baryshnikov, and **William Handley**. AEONS: approximating the end of nested sampling. *MNRAS*, 532(4):4035–4049, August 2024.
[37] T. Gessey-Jones, S. Pochinda, H. T. J. Bevins, A. Fialkov, **W. J. Handley**, E. de Lera Acedo, S. Singh, and R. Barkana. On the constraints on superconducting cosmic strings from 21-cm cosmology. *MNRAS*, 529(1):519–536, March 2024.
[38] T. Gessey-Jones and **W. J. Handley**. Fully Bayesian forecasts with evidence networks. *PRD*, 109(12):123541, June 2024.
[39] D. D. Dineen and **W. J. Handley**. Analytic approximations for the primordial power spectrum with Israel junction conditions. *PRD*, 109(8):083513, April 2024.
[40] Wei-Ning Deng and **Will Handley**. Predicting spatial curvature Ω_K in globally CPT-symmetric universes. *arXiv*, 2407.18225, July 2024.
[41] Wei-Ning Deng and **Will Handley**. Predicting spatial curvature ΩK in globally CPT-symmetric universes. *PRD*, 110(10):103528, November 2024.
[42] Harry T. J. Bevins, Stefan Heimersheim, Irene Abril-Cabezas, Anastasia Fialkov, Eloy de Lera Acedo, **William Handley**, Saurabh Singh, and Rennan Barkana. Joint analysis constraints on the physics of the first galaxies with low-frequency radio astronomy data. *MNRAS*, 527(1):813–827, January 2024.
[43] David Yallup and **Will Handley**. Hunting for bumps in the margins. *Journal of Instrumentation*, 18(5):P05014, May 2023.
[44] Margret Westerkamp, Jakob Roth, Philipp Frank, **Will Handley**, and Torsten Enßlin. Inferring Evidence from Nested Sampling Data via Information Field Theory. *arXiv*, 2312.11907, December 2023.
[45] K. H. Scheutwinkel, **W. Handley**, and E. de Lera Acedo. Bayesian evidence-driven likelihood selection for sky-averaged 21-cm signal extraction. *PASA*, 40:e016, April 2023.
[46] Anchal Saxena, P. Daniel Meerburg, Eloy de Lera Acedo, **Will Handley**, and Léon V. E. Koopmans. Sky-averaged 21-cm signal extraction using multiple antennas with an SVD framework: the REACH case. *MNRAS*, 522(1):1022–1032, June 2023.
[47] A. N. Ormondroyd, **W. J. Handley**, M. P. Hobson, and A. N. Lasenby. Balancing ACT: weighing prior dependency and global tensions of DR6 lensing with other datasets. *arXiv*, 2310.08490, October 2023.
[48] S. A. K. Leeney, **W. J. Handley**, and E. de Lera Acedo. Bayesian approach to radio frequency interference mitigation. *PRD*, 108(6):062006, September 2023.
[49] T. Gessey-Jones, A. Fialkov, E. de Lera Acedo, **W. J. Handley**, and R. Barkana. Signatures of cosmic ray heating in 21-cm observables. *MNRAS*, 526(3):4262–4284, December 2023.
[50] et al. and Gábor Csányi. A foundation model for atomistic materials chemistry. *arXiv*, 2401.00096, December 2023.
[51] Harry T. J. Bevins, **William J. Handley**, Pablo Lemos, Peter H. Sims, Eloy de Lera Acedo, Anastasia Fialkov, and Justin Alsing. Marginal post-processing of Bayesian inference products with normalizing flows and kernel density estimators. *MNRAS*, 526(3):4613–4626, December 2023.
[52] Harry Bevins, **Will Handley**, and Thomas Gessey-Jones. Piecewise Normalizing Flows. *arXiv*, 2305.02930, May 2023.
[53] Dominic Anstey, Eloy de Lera Acedo, and **Will Handley**. Use of time dependent data in Bayesian global 21-cm foreground and signal modelling. *MNRAS*, 520(1):850–865, March 2023.
[54] David Yallup, Timo Janßen, Steffen Schumann, and **Will Handley**. Exploring phase space with nested sampling. *European Physical Journal C*, 82(8):678, August 2022.
[55] David Yallup, **Will Handley**, Mike Hobson, Anthony Lasenby, and Pablo Lemos. Split personalities in Bayesian Neural Networks: the case for full marginalisation. *arXiv*, 2205.11151, May 2022.
[56] Zakhar Shumaylov and **Will Handley**. Primordial power spectra from k -inflation with curvature. *PRD*, 105(12):123532, June 2022.

[57] K. H. Scheutwinkel, E. de Lera Acedo, and **W. Handley**. Bayesian evidence-driven diagnosis of instrumental systematics for sky-averaged 21-cm cosmology experiments. *PASA*, 39:e052, October 2022.
[58] REACH collaboration. The REACH radiometer for detecting the 21-cm hydrogen signal from redshift z ≈ 7.5-28. *Nature Astronomy*, 6:984–998, July 2022.
[59] REACH collaboration. Radio Antenna Design for Sky-Averaged 21cm Cosmology Experiments: The REACH Case. *Journal of Astronomical Instrumentation*, 11(1):2250001–2058, January 2022.
[60] Metha Prathaban and **Will Handley**. Rescuing palindromic universes with improved recombination modeling. *PRD*, 105(12):123508, June 2022.
[61] Aleksandr Petrosyan and **William James Handley**. SuperNest: accelerated nested sampling applied to astrophysics and cosmology. *arXiv*, 2212.01760, December 2022.
[62] A. N. Lasenby, **W. J. Handley**, D. J. Bartlett, and C. S. Negreanu. Perturbations and the future conformal boundary. *PRD*, 105(8):083514, April 2022.
[63] L. T. Hergt, F. J. Agocs, **W. J. Handley**, M. P. Hobson, and A. N. Lasenby. Finite inflation in curved space. *PRD*, 106(6):063529, September 2022.
[64] T. Gessey-Jones, N. S. Sartorio, A. Fialkov, G. M. Mirouh, M. Magg, R. G. Izzard, E. de Lera Acedo, **W. J. Handley**, and R. Barkana. Impact of the primordial stellar initial mass function on the 21-cm signal. *MNRAS*, 516(1):841–860, October 2022.
[65] Gambit Collaboration and et al. Simple and statistically sound recommendations for analysing physical theories. *Reports on Progress in Physics*, 85(5):052201, May 2022.
[66] Andrew Fowlie, Sebastian Hoof, and **Will Handley**. Nested Sampling for Frequentist Computation: Fast Estimation of Small p -Values. *PRL*, 128(2):021801, January 2022.
[67] Harry Bevins, **Will Handley**, Pablo Lemos, Peter Sims, Eloy de Lera Acedo, and Anastasia Fialkov. Marginal Bayesian Statistics Using Masked Autoregressive Flows and Kernel Density Estimators with Examples in Cosmology. *arXiv*, 2207.11457, July 2022.
[68] H. T. J. Bevins, A. Fialkov, E. de Lera Acedo, **W. J. Handley**, S. Singh, R. Subrahmanyan, and R. Barkana. Astrophysical constraints from the SARAS 3 non-detection of the cosmic dawn sky-averaged 21-cm signal. *Nature Astronomy*, 6:1473–1483, December 2022.
[69] H. T. J. Bevins, E. de Lera Acedo, A. Fialkov, **W. J. Handley**, S. Singh, R. Subrahmanyan, and R. Barkana. A comprehensive Bayesian reanalysis of the SARAS2 data from the epoch of reionization. *MNRAS*, 513(3):4507–4526, July 2022.
[70] D. J. Bartlett, **W. J. Handley**, and A. N. Lasenby. Improved cosmological fits with quantized primordial power spectra. *PRD*, 105(8):083515, April 2022.
[71] Csaba Balázs, Sanjay Bloor, Tomás E. Gonzalo, **Will Handley**, Sebastian Hoof, Felix Kahlhoefer, Marie Lecroq, David J. E. Marsh, Janina J. Renk, Pat Scott, and Patrick Stöcker. Cosmological constraints on decaying axion-like particles: a global analysis. *JCAP*, 2022(12):027, December 2022.
[72] Greg Ashton, Noam Bernstein, Johannes Buchner, Xi Chen, Gábor Csányi, Andrew Fowlie, Farhan Feroz, Matthew Griffiths, **Will Handley**, Michael Habeck, Edward Higson, Michael Hobson, Anthony Lasenby, David Parkinson, Livia B. Pártay, Matthew Pitkin, Doris Schneider, Joshua S. Speagle, Leah South, John Veitch, Philipp Wacker, David J. Wales, and David Yallup. Nested sampling for physical scientists. *Nature Reviews Methods Primers*, 2:39, May 2022.
[73] Dominic Anstey, John Cumner, Eloy de Lera Acedo, and **Will Handley**. Informing antenna design for sky-averaged 21-cm experiments using a simulated Bayesian data analysis pipeline. *MNRAS*, 509(4):4679–4693, February 2022.
[74] Ayngaran Thavanesan, Denis Werth, and **Will Handley**. Analytical approximations for curved primordial power spectra. *PRD*, 103(2):023519, January 2021.
[75] Emma Shen, Dominic Anstey, Eloy de Lera Acedo, Anastasia Fialkov, and **Will Handley**. Quantifying ionospheric effects on global 21-cm observations. *MNRAS*, 503(1):344–353, May 2021.
[76] I. L. V. Roque, **W. J. Handley**, and N. Razavi-Ghods. Bayesian noise wave calibration for 21-cm global experiments. *MNRAS*, 505(2):2638–2646, August 2021.
[77] B. Joachimi, F. Köhlinger, **W. Handley**, and P. Lemos. When tension is just a fluctuation. How noisy data affect model comparison. *A&A*, 647:L5, March 2021.
[78] L. T. Hergt, **W. J. Handley**, M. P. Hobson, and A. N. Lasenby. Bayesian evidence for the tensor-to-scalar ratio r and neutrino masses m_ν : Effects of uniform versus logarithmic priors. *PRD*, 103(12):123511, June 2021.
[79] **Will Handley** and Pablo Lemos. Quantifying the global parameter tensions between ACT, SPT, and Planck. *PRD*, 103(6):063529, March 2021.
[80] **Will Handley**. Curvature tension: Evidence for a closed universe. *PRD*, 103(4):L041301, February 2021.
[81] W. I. J. Haddadin and **W. J. Handley**. Rapid numerical solutions for the Mukhanov-Sasaki equation. *PRD*, 103(12):123513, June 2021.
[82] T. Gessey-Jones and **W. J. Handley**. Constraining quantum initial conditions before inflation. *PRD*, 104(6):063532, September 2021.
[83] Gambit Cosmology Workgroup. Strengthening the bound on the mass of the lightest neutrino with terrestrial and cosmological experiments. *PRD*, 103(12):123508, June 2021.
[84] Gambit Cosmology Workgroup. CosmoBit: a GAMBIT module for computing cosmological observables and likelihoods. *JCAP*, 2021(2):022, February 2021.
[85] Gambit Collaboration. Thermal WIMPs and the scale of new physics: global fits of Dirac dark matter effective field theories. *European Physical Journal C*, 81(11):992, November 2021.

[86] Andrew Fowlie, **Will Handley**, and Liangliang Su. Nested sampling with plateaus. *MNRAS*, 503(1):1199–1205, May 2021.
[87] DarkMachines High Dimensional Sampling Group. A comparison of optimisation algorithms for high-dimensional particle and astrophysics applications. *Journal of High Energy Physics*, 2021(5):108, May 2021.
[88] Ethan Carragher, **Will Handley**, Daniel Murnane, Peter Stangl, Wei Su, Martin White, and Anthony G. Williams. Convergent Bayesian global fits of 4D composite Higgs models. *Journal of High Energy Physics*, 2021(5):237, May 2021.
[89] H. T. J. Bevins, **W. J. Handley**, A. Fialkov, E. de Lera Acedo, and K. Javid. GLOBALEMU: a novel and robust approach for emulating the sky-averaged 21-cm signal from the cosmic dawn and epoch of reionization. *MNRAS*, 508(2):2923–2936, December 2021.
[90] H. T. J. Bevins, **W. J. Handley**, A. Fialkov, E. de Lera Acedo, L. J. Greenhill, and D. C. Price. MAXSMOOTH: rapid maximally smooth function fitting with applications in Global 21-cm cosmology. *MNRAS*, 502(3):4405–4425, April 2021.
[91] W. E. V. Barker, A. N. Lasenby, M. P. Hobson, and **W. J. Handley**. Nonlinear Hamiltonian analysis of new quadratic torsion theories: Cases with curvature-free constraints. *PRD*, 104(8):084036, October 2021.
[92] Dominic Anstey, Eloy de Lera Acedo, and **Will Handley**. A general Bayesian framework for foreground modelling and chromaticity correction for global 21 cm experiments. *MNRAS*, 506(2):2041–2058, September 2021.
[93] Justin Alsing and **Will Handley**. Nested sampling with any prior you like. *MNRAS*, 505(1):L95–L99, July 2021.
[94] E. Ahrer, D. Queloz, V. M. Rajpaul, D. Ségransan, F. Bouchy, R. Hall, **W. Handley**, C. Lovis, M. Mayor, A. Mortier, F. Pepe, S. Thompson, S. Udry, and N. Unger. The HARPS search for southern extra-solar planets - XLV. Two Neptune mass planets orbiting HD 13808: a study of stellar activity modelling's impact on planet detection. *MNRAS*, 503(1):1248–1263, May 2021.
[95] F. Lienhard, D. Queloz, M. Gillon, A. Burdanov, L. Delrez, E. Ducrot, **W. Handley**, E. Jehin, C. A. Murray, A. H. M. J. Triaud, E. Gillen, A. Mortier, and B. V. Rackham. Global analysis of the TRAPPIST Ultra-Cool Dwarf Transit Survey. *MNRAS*, 497(3):3790–3808, September 2020.
[96] Pablo Lemos, Fabian Köhlinger, **Will Handley**, Benjamin Joachimi, Lorne Whiteway, and Ofer Lahav. Quantifying Suspiciousness within correlated data sets. *MNRAS*, 496(4):4647–4653, August 2020.
[97] Kamran Javid, **Will Handley**, Mike Hobson, and Anthony Lasenby. Compromise-free Bayesian neural networks. *arXiv*, 2004.12211, April 2020.
[98] Andrew Fowlie, **Will Handley**, and Liangliang Su. Nested sampling cross-checks using order statistics. *MNRAS*, 497(4):5256–5263, October 2020.
[99] W. E. V. Barker, A. N. Lasenby, M. P. Hobson, and **W. J. Handley**. Systematic study of background cosmology in unitary Poincaré gauge theories with application to emergent dark radiation and H_0 tension. *PRD*, 102(2):024048, July 2020.
[100] W. E. V. Barker, A. N. Lasenby, M. P. Hobson, and **W. J. Handley**. Mapping Poincaré gauge cosmology to Horndeski theory for emergent dark energy. *PRD*, 102(8):084002, October 2020.
[101] W. E. V. Barker, A. N. Lasenby, M. P. Hobson, and **W. J. Handley**. Addressing H_0 tension with emergent dark radiation in unitary gravity. *arXiv*, 2003.02690, March 2020.
[102] Jamie Bamber and **Will Handley**. Beyond the Runge-Kutta-Wentzel-Kramers-Brillouin method. *PRD*, 101(4):043517, February 2020.
[103] F. J. Agocs, M. P. Hobson, **W. J. Handley**, and A. N. Lasenby. Dense output for highly oscillatory numerical solutions. *arXiv*, 2007.05013, July 2020.
[104] F. J. Agocs, L. T. Hergt, **W. J. Handley**, A. N. Lasenby, and M. P. Hobson. Quantum initial conditions for inflation and canonical invariance. *PRD*, 102(2):023507, July 2020.
[105] F. J. Agocs, **W. J. Handley**, A. N. Lasenby, and M. P. Hobson. Efficient method for solving highly oscillatory ordinary differential equations with applications to physical systems. *Physical Review Research*, 2(1):013030, January 2020.
[106] James S. Spencer, Nick S. Blunt, Seonghoon Choi, Jiri Etrych, Maria-Andreea Filip, W. M. C. Foulkes, Ruth S. T. Franklin, **Will J. Handley**, Fionn D. Malone, Verena A. Neufeld, Roberto Di Remigio, Thomas W. Rogers, Charles J. C. Scott, James J. Shepherd, William A. Vigor, Joseph Weston, RuQing Xu, and Alex J. W. Thom. The HANDE-QMC project: open-source stochastic quantum chemistry from the ground state up. *Journal of Chemical Theory and Computation*, 15(3):1728–1742, January 2019.
[107] Edward Higson, **Will Handley**, Mike Hobson, and Anthony Lasenby. Dynamic nested sampling: an improved algorithm for parameter estimation and evidence calculation. *Statistics and Computing*, 29(5):891–913, September 2019.
[108] Edward Higson, **Will Handley**, Michael Hobson, and Anthony Lasenby. NESTCHECK: diagnostic tests for nested sampling calculations. *MNRAS*, 483(2):2044–2056, February 2019.
[109] Edward Higson, **Will Handley**, Michael Hobson, and Anthony Lasenby. Bayesian sparse reconstruction: a brute-force approach to astronomical imaging and machine learning. *MNRAS*, 483(4):4828–4846, March 2019.
[110] L. T. Hergt, **W. J. Handley**, M. P. Hobson, and A. N. Lasenby. Constraining the kinetically dominated universe. *PRD*, 100(2):023501, July 2019.
[111] L. T. Hergt, **W. J. Handley**, M. P. Hobson, and A. N. Lasenby. Case for kinetically dominated initial conditions for inflation. *PRD*, 100(2):023502, July 2019.
[112] **Will J. Handley**, Anthony N. Lasenby, Hiranya V. Peiris, and Michael P. Hobson. Bayesian inflationary reconstructions from Planck 2018 data. *PRD*, 100(10):103511, November 2019.

[113] **Will Handley** and Marius Millea. Maximum-Entropy Priors with Derived Parameters in a Specified Distribution. *Entropy*, 21(3):272, March 2019.
[114] **Will Handley** and Pablo Lemos. Quantifying tensions in cosmological parameters: Interpreting the DES evidence ratio. *PRD*, 100(4):043504, August 2019.
[115] **Will Handley** and Pablo Lemos. Quantifying dimensionality: Bayesian cosmological model complexities. *PRD*, 100(2):023512, July 2019.
[116] **Will Handley**, Anthony Lasenby, and Mike Hobson. Logolinear series expansions with applications to primordial cosmology. *PRD*, 99(12):123512, June 2019.
[117] **Will Handley**. Primordial power spectra for curved inflating universes. *PRD*, 100(12):123517, July 2019.
[118] **Will Handley**. anesthetic: nested sampling visualisation. *JOSS*, 4:1414, May 2019.
[119] W. E. V. Barker, A. N. Lasenby, M. P. Hobson, and **W. J. Handley**. Static energetics in gravity. *JMAP*, 60(5):052504, May 2019.
[120] Edward Higson, **Will Handley**, Mike Hobson, and Anthony Lasenby. Sampling Errors in Nested Sampling Parameter Estimation. *Bayesian Analysis*, 13(3):873–896, March 2018.
[121] **Will Handley**. fgivenx: A Python package for functional posterior plotting. *JOSS*, 3(28):849, August 2018.
[122] Richard D. Hall, Samantha J. Thompson, **Will Handley**, and Didier Queloz. On the Feasibility of Intense Radial Velocity Surveys for Earth-Twin Discoveries. *MNRAS*, 479(3):2968–2987, September 2018.
[123] A. J. K. Chua, S. Hee, **W. J. Handley**, E. Higson, C. J. Moore, J. R. Gair, M. P. Hobson, and A. N. Lasenby. Towards a framework for testing general relativity with extreme-mass-ratio-inspiral observations. *MNRAS*, 478(1):28–40, July 2018.
[124] Gong-Bo Zhao, Marco Raveri, Levon Pogosian, Yuting Wang, Robert G. Crittenden, **Will J. Handley**, and et al. Dynamical dark energy in light of the latest observations. *Nature Astronomy*, 1:627–632, August 2017.
[125] S. Hee, J. A. Vázquez, **W. J. Handley**, M. P. Hobson, and A. N. Lasenby. Constraining the dark energy equation of state using Bayes theorem and the Kullback-Leibler divergence. *MNRAS*, 466(1):369–377, April 2017.
[126] Clare Rumsey, Malak Olamaie, Yvette C. Perrott, Helen R. Russell, Farhan Feroz, Keith J. B. Grainge, **Will J. Handley**, Michael P. Hobson, Richard D. E. Saunders, and Michel P. Schammel. AMI observations of 10 CLASH galaxy clusters: SZ and X-ray data used together to determine cluster dynamical states. *MNRAS*, 460(1):569–589, July 2016.
[127] S. Hee, **W. J. Handley**, M. P. Hobson, and A. N. Lasenby. Bayesian model selection without evidences: application to the dark energy equation-of-state. *MNRAS*, 455(3):2461–2473, January 2016.
[128] **W. J. Handley**, A. N. Lasenby, and M. P. Hobson. The Runge-Kutta-Wentzel-Kramers-Brillouin Method. *arXiv*, 1612.02288, December 2016.
[129] **W. J. Handley**, A. N. Lasenby, and M. P. Hobson. Novel quantum initial conditions for inflation. *PRD*, 94(2):024041, July 2016.
[130] **W. J. Handley**, M. P. Hobson, and A. N. Lasenby. POLYCHORD: next-generation nested sampling. *MNRAS*, 453(4):4384–4398, November 2015.
[131] **W. J. Handley**, M. P. Hobson, and A. N. Lasenby. polychord: nested sampling for cosmology. *MNRAS*, 450:L61–L65, June 2015.
[132] **W. J. Handley**, S. D. Brechet, A. N. Lasenby, and M. P. Hobson. Kinetic initial conditions for inflation. *PRD*, 89(6):063505, March 2014.

CORE Collaboration

[133] Exploring cosmic origins with CORE: The instrument. *JCAP*, 2018(4):015, April 2018.
[134] Exploring cosmic origins with CORE: Survey requirements and mission design. *JCAP*, 2018(4):014, April 2018.
[135] Exploring cosmic origins with CORE: Mitigation of systematic effects. *JCAP*, 2018(4):022, April 2018.
[136] Exploring cosmic origins with CORE: Inflation. *JCAP*, 2018(4):016, April 2018.
[137] Exploring cosmic origins with CORE: Gravitational lensing of the CMB. *JCAP*, 2018(4):018, April 2018.
[138] Exploring cosmic origins with CORE: Extragalactic sources in cosmic microwave background maps. *JCAP*, 2018(4):020, April 2018.
[139] Exploring cosmic origins with CORE: Effects of observer peculiar motion. *JCAP*, 2018(4):021, April 2018.
[140] Exploring cosmic origins with CORE: Cosmological parameters. *JCAP*, 2018(4):017, April 2018.
[141] Exploring cosmic origins with CORE: Cluster science. *JCAP*, 2018(4):019, April 2018.
[142] Exploring cosmic origins with CORE: B-mode component separation. *JCAP*, 2018(4):023, April 2018.

Planck Collaboration

[143] Planck intermediate results. LVII. Joint Planck LFI and HFI data processing. *A&A*, 643:A42, November 2020.
[144] Planck intermediate results. LVI. Detection of the CMB dipole through modulation of the thermal Sunyaev-Zeldovich effect: Eppur si muove II. *A&A*, 644:A100, December 2020.
[145] Planck intermediate results. LV. Reliability and thermal properties of high-frequency sources in the Second Planck Catalogue of Compact Sources. *A&A*, 644:A99, December 2020.
[146] Planck 2018 results. XII. Galactic astrophysics using polarized dust emission. *A&A*, 641:A12, September 2020.
[147] Planck 2018 results. XI. Polarized dust foregrounds. *A&A*, 641:A11, September 2020.
[148] Planck 2018 results. X. Constraints on inflation. *A&A*, 641:A10, September 2020.
[149] Planck 2018 results. VIII. Gravitational lensing. *A&A*, 641:A8, September 2020.
[150] Planck 2018 results. VII. Isotropy and statistics of the CMB. *A&A*, 641:A7, September 2020.
[151] Planck 2018 results. VI. Cosmological parameters. *A&A*, 641:A6, September 2020.
[152] Planck 2018 results. V. CMB power spectra and likelihoods. *A&A*, 641:A5, September 2020.
[153] Planck 2018 results. IX. Constraints on primordial non-Gaussianity. *A&A*, 641:A9, September 2020.

[154] Planck 2018 results. IV. Diffuse component separation. *A&A*, 641:A4, September 2020.
[155] Planck 2018 results. III. High Frequency Instrument data processing and frequency maps. *A&A*, 641:A3, September 2020.
[156] Planck 2018 results. II. Low Frequency Instrument data processing. *A&A*, 641:A2, September 2020.
[157] Planck 2018 results. I. Overview and the cosmological legacy of Planck. *A&A*, 641:A1, September 2020.
[158] Planck intermediate results. LIV. The Planck multi-frequency catalogue of non-thermal sources. *A&A*, 619:A94, November 2018.
[159] Planck intermediate results. LIII. Detection of velocity dispersion from the kinetic Sunyaev-Zeldovich effect. *A&A*, 617:A48, September 2018.
[160] Planck 2015 results. XX. Constraints on inflation. *A&A*, 594:A20, September 2016.
[161] Planck 2015 results. I. Overview of products and scientific results. *A&A*, 594:A1, September 2016.